MAT 201 - MATHEMATICAL METHODS I

LECTURE 9: EVALUATION OF LINE INTEGRALS

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OUTLINE

- Introduction
- Alternative form of a line integral
- Properties of line integrals
- Line integral around a closed curve
- Line integral in space
- Exact differential
- Dependence of line integral on the path of integration
- Summary

0:04

INTRODUCTION

If a field exists in the $x - y$ plane, producing a force F on a particle at the point K, then F can be resolved into two components:

- F_t along the tangent to the curve at K
- F_n along the normal to the curve at K

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INTRODUCTION

The work done in moving the particle through a small distance δs from K to L along the curve is then approximately $F_t \delta s$. So the total work done in moving a particle along the curve from A to B is given by

$$\lim_{\delta s \to 0} \sum F_t \delta s = \int F_t ds, \quad \text{from A to B} \tag{1}$$

This is normally written as $\int_{AB} F_t ds$ where A and B are the end points of the curve, or as $\int_c F_t ds$, where the curve, c, connecting A and B is defined.

LINE INTEGRAL [DEFINITION]

Such an integral thus formed is called a ***line integral*** since integration is carried out along the path of the particular curve c joining A and B.

$$\int_{AB} F_t ds = \int_c F_t ds \tag{2}$$

where c is the curve $y = f(x)$ between $A(x, y)$ and $B(x, y)$.

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ALTERNATIVE FORM OF A LINE INTEGRAL

It is often convenient to integrate with respect to x or y than to take arc length as the variable. If F_t has a component P in the $x-$direction and Q in the y-direction, then the work done from K to L can be stated as $P\delta x + Q\delta y$. Therefore

$$I = \int_{AB} F_t ds = \int_{AB} (Pdx + Qdy) \tag{3}$$

where c is the prescribed curve and F_t, P and Q are functions of x and y.

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EXAMPLE 1

Evaluate $I = \int_c \left\{ \left(x^2 + 2y\right) dx + xydy \right\}$ from $O(0,0)$ to $B(1,4)$ along the curve $y = 4x^2$.
In this case, c is the curve $y = 4x^2$, therefore $dy = 8xdx$.
Now, $I = \int_0^1 \left(9x^2 + 32x^4\right) dx = 9.4$

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PROPERTIES OF LINE INTEGRALS

- $\int_c Fds = \int_c (Pdx + Qdy)$
- $\int_{AB} Fds = -\int_{BA} Fds$ and $\int_{AB} (Pdx + Qdy) = -\int_{BA} (Pdx + Qdy)$
- If the path of integration c joining A to B is divided into two parts AK and KB, then $I_c = I_{AB} = I_{AK} + I_{KB}$
- In all cases, the function $y(x)$ that describes the path of integration involved must be continuous and single-valued
- If the function $y(x)$ that describes the path of integration c, is not single-valued, then the path is divided into two or more in such a way that the resulting sub-paths are single-valued.

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EXAMPLE 2

Evaluate $I = \int_c \{(x+y)\,dx\}$ from $O(0,1)$ to $B(0,-1)$ along the semi-circle $x^2+y^2=1$ for $x \geq 0$.

For any value of x, $y = \pm(1-x^2)^{1/2}$. Therefore, we divide c into two parts:

- $y = (1-x^2)^{1/2}$ from A to K
- $y = -(1-x^2)^{1/2}$ from K to B

0:04

EXAMPLE 2 [CONT.]

Now, $I = \int_c (Pdx + Qdy)$, but $Q = 0$ in this case, therefore

$$I = \int_c Pdx \tag{4}$$

$$= \int_0^1 \left(x + \left(1 - x^2\right)^{\frac{1}{2}}\right) dx + \int_1^0 \left(x - \left(1 - x^2\right)^{\frac{1}{2}}\right) dx \tag{5}$$

$$= 2\int_0^1 \left(1 - x^2\right)^{\frac{1}{2}} dx = \int_0^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta = \frac{\pi}{2} \tag{6}$$

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SIMPLY CONNECTED REGION [DEFINITION]

A region is said to be simply connected if a path joining A and B can be deformed to coincide with any other line joining A and B without going outside the region. In other words, a region is simply connected if any closed path in the region can be contracted to a single point without leaving the region.

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LINE INTEGRAL AROUND A CLOSED CURVE

An integral to be evaluated around a closed curve in the positive (counter-clockwise) direction is usually denoted by the symbol

$$\oint$$

With a closed curve, the y-values on the path c, cannot be single-valued, therefore, we divide the path into two or more paths and treat each separately.

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EXAMPLE 3

Evaluate the line integral $I = \oint_c \left\{x^2 dx - 2xydy\right\}$, where c comprises the three sides of the triangle joining $O(0,0)$, $A(1,0)$ and $B(0,1)$.

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EXAMPLE 4

Evaluate $I = \oint_c \left\{ xydx + (1+y^2)dy \right\}$, where c is the boundary of the rectangle joining $A(1,0)$, $B(3,0)$, $C(3,2)$ and $D(1,2)$.

0:04

LINE INTEGRAL IN SPACE

Suppose that $\mathcal{C}$ is a smooth space curve given by the parametric equations:
$x = x(t), \quad y = y(t) \quad \text{and} \quad z = z(t), \qquad t \in [a, b]$
or by a vector equation $r(t) = x(t)i + y(t)j + z(t)k$. If f is a function of three variables that is continuous on some region containing $\mathcal{C}$, then we define the line integral of f along $\mathcal{C}$ as

$$\int_c f(x, y, z)ds = \int_a^b f\left(x(t), y(t), z(t)\right) \times \left(\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right)^{\frac{1}{2}} dt \tag{7}$$

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EXAMPLE 5

Evaluate $\int_c y \sin z ds$, where c is the circular helix given by the equations: $x = \cos t$, $y = \sin t, z = t, 0 \leq t \leq 2\pi$.

$$\int_c y \sin z ds = \int_0^{2\pi} \left(\sin^2 t\right) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \tag{8}$$

$$= \int_0^{2\pi} \left(\sin^2 t\right) \sqrt{\sin^2 t + \cos^2 t + 1}\ dt \tag{9}$$

$$= \sqrt{2} \int_0^{2\pi} \frac{1}{2}(1 - \cos 2t) dt = \sqrt{2}\pi \tag{10}$$

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EXACT DIFFERENTIAL

If $g = Pdx + Qdy$ is exact, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \tag{11}$$

If $g = Pdx + Qdy + Rdz$ is exact, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}; \quad \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}; \quad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y} \tag{12}$$

where P, Q and their first derivatives are finite and continuous inside the simply connected region $\mathcal{R}$

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DEPENDENCE OF LINE INTEGRAL ON THE PATH OF INTEGRATION

If g is an exact differential, then $\int_c g$ is

- Independent of the path of integration where c lies entirely within $\mathcal{R}$.
- Zero when c is a closed curve lying within $\mathcal{R}$.

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EXAMPLE 6

Show that $I = \int_c \left\{ \left(3x^2 \sin y + 2\sin 2x + y^3\right) dx + \left(x^3 \cos y + 3xy^2\right) dy \right\}$ is independent of the path of integration and evaluate it from $A(0,0)$ to $B(\pi/2, \pi)$.

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SUMMARY

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